

unusable. There are three basic kinds of lights in Blender (and most of the other modelling software as well). They are:

1. Directional Light

When the light source is far away (say the sun), the light rays coming from it are almost parallel to each other. It almost looks like the light rays are coming from the same direction. Such a light which is situated at an infinitely large distance (for practical purposes) is called a Directional Light. A directional light is just represented by a direction in space (It does not have a position). Directional lights do not attenuate, and their intensities remain constant through their propagation.

Essentially, this is the light you need to simulate the Sun. In Blender you can create a directional light by either setting the light type as “Sun” or another way to do this is to create a plane and set its emission properties. A plane emitting light is almost equivalent to a directional light source (the only difference is that its emission does attenuate).

2. Point Lights

A Point Light is a point in space which sends light rays originating from the source in all directions around it. It is just represented by a point in the space. The light coming from a point light attenuates in intensity by the inverse of the quadratic rule (you can tweak this as per your liking in the light settings). The attenuation factor is given by the following formula, where the constants K_c , K_l , K_q are the factors for the constant, linear, and quadratic terms (usually, K_c is kept one, K_l quite small and K_q even smaller):

$$1.0$$

$$I_{F_{att}} = K_c + K_l * d + K_q * d^2$$

Directional Lights are great as global light sources, but to illuminate small nicks and corners of your scenes, you need to have point lights as well. Usually, a good scene has point lights scattered throughout. Since a single point light can cause long and unending shadows, it is usually a good practice to have a multitude of them at various locations.

3. Spot Lights

The last kind of light Blender provides a Spot Light. Think of a spot light as a torch, which has a central region in which the light is focused, and outside which it attenuates fast. It is represented by a point and a direction